Adding Text to Drawings

CADD allows you to add fine lettering to your drawings. You can use text to write notes, specifications and to describe the components of a drawing. Text created with CADD is neat, stylish and can be easily edited. Typing skills are helpful if you intend to write a lot of text.

Writing text with CADD is as simple as typing it on the keyboard. You can locate it anywhere on the drawing, write it as big or as small as you like and choose from a number of available fonts.

Note:

When large amounts of text are added to drawings, it slows down the screen displays. Many programs provide options to temporarily turn off text or to display text outlines only. This feature helps save computer memory and speeds up the display of screen images. The text can be turned back on whenever needed.

The following are the basic factors that control the appearance of text:
(The exact terms and procedures used vary from one program to another.)

- Text height
- Height to width ratio and inclination of letters
- Special effects
- Alignment of text (justification)
- Text fonts

Text Height

You can write any size text by specifying the height of letters. It is a good idea to develop a set of standards determining what size text to use where. For example, you can use 1/16" or 3/32" text height for descriptive text, 1/8" height for sub-titles and 1/4" for main titles (Diagram A, Fig. 3.9). The height standard you choose for text in one drawing should be followed on all drawings in a project. This presents a uniform look to each drawing in the project.

Note:

The height of text you use is reduced or enlarged depending on the scale factor applied to drawings when they are plotted. Accordingly, you need to enter smaller or bigger text. We will discuss more on this in Chapter 8, "Printing and Plotting."

Height to Width Ratio and Inclination of Letters

CADD allows you to fine-tune the text in a number of ways. You can specify a height-to-width ratio for all the letters, indicate spacing between the letters and even make the letters inclined.

Diagram B (Fig. 3.9) illustrates how the text appearance is altered with a different height to width ratio. Diagram C (Fig. 3.9) illustrates how you can change the appearance of text by inclining it to a specific angle.

Special Effects

Sometimes you need to write text rotated at an angle, written in a vertical direction, or even mirrored. Many CADD programs provide these special options that can be used for specific drawing conditions. Diagram D (Fig. 3.9) illustrates some examples of drawing rotated, mirrored and vertical text.

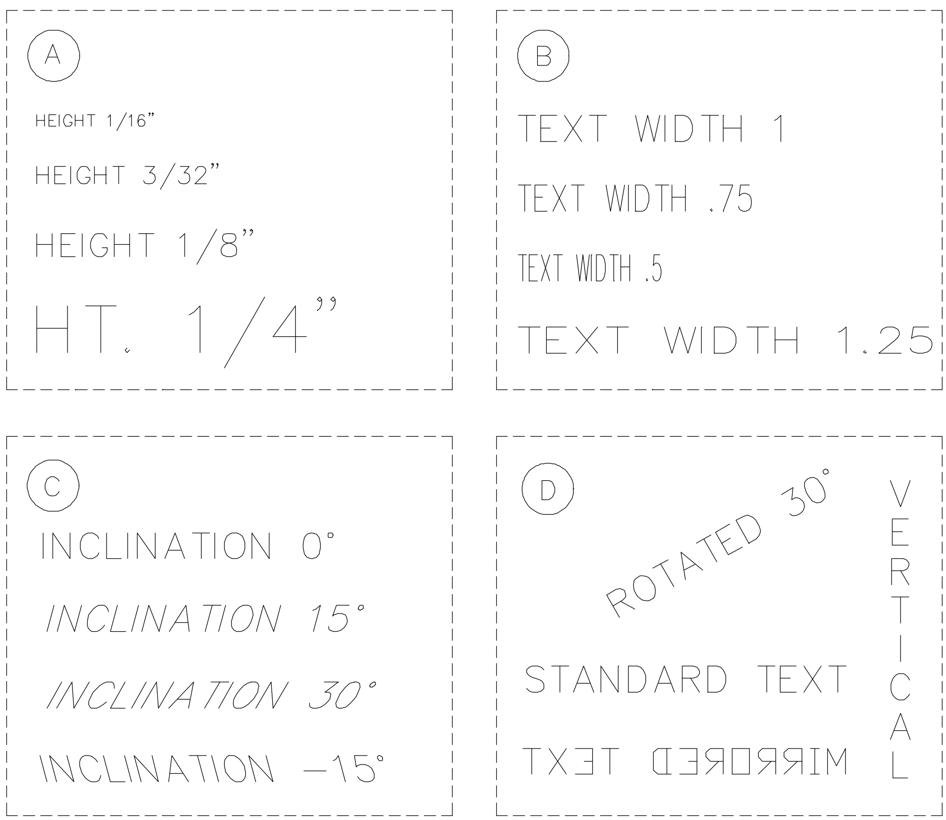

Fig. 3.9: *Drawing text with varying height, ratio, spacing and inclination.*

Alignment of Text (Justification)

The term justification is commonly referred to as the alignment of text. The text can be aligned to the left (left justified), aligned to the right (right justified) or centered (center justified). Fig. 3.10 illustrates examples of text written with left, centered or right justification. You can specify justification for individual lines or paragraphs.

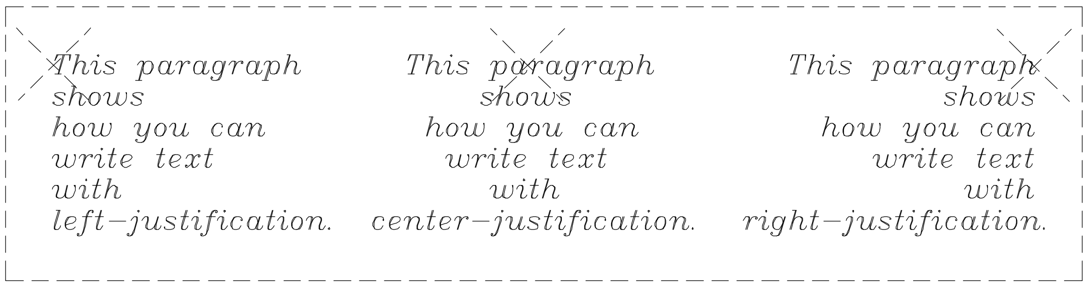

Fig. 3.10: *Drawing text with justification.*

Text Fonts

CADD offers many fonts that allow you to write text with different styles. You can choose from dozens of fonts available in CADD. A number of fonts are also available through independent vendors. Fig. 3.11 illustrates some of the common fonts. You can give your drawing whatever presentation you want by selecting an appropriate font.

Fig. 3.11: *Drawing text using different fonts.*

Notes:

- Most CADD developers include proprietary fonts with their software. Font names change from one program to another.
- The fonts used in a drawing must be supported by the printer or plotter used for printing the drawing. You may want to do a test plot before using specific fonts in a drawing.
- When sending drawings electronically, make sure to include the fonts used in the drawing. Otherwise, the recipient may not be able to open the drawings.
- You can change the appearance of text using the editing functions of CADD. You can change the height, width, inclination, rotation, spacing, fonts, justification, etc., as needed.

Defining a Text Style

As discussed, there are a number of factors that control the appearance of text. It is time-consuming to specify every parameter each time you need to write text. CADD allows you to define text styles that contain all the text information such as size, justification and font. When you need to write text, simply select a particular style and all the text thereafter is written with that style. CADD offers a number of ready-made text styles as well.

Important Tip:

There are a number of add-on programs available that can make working with text faster and easier. These programs provide basic word-processing capabilities that can be used to write reports and make charts. They provide access to a dictionary and thesaurus database that can be used to check spelling and to search for alternative words.

Drawing Dimensions

CADD's dimensioning functions provide a fast and accurate means for drawing dimensions. To draw a dimension, all you need to do is to indicate the points that need to be dimensioned. CADD automatically calculates the dimension value and draws all the necessary annotations.

Fig. 3.12 illustrates the basic components of a CADD dimension. The annotations that form a dimension are: dimension line, dimension text, dimension terminators and extension lines. You can control the appearance of each of these elements by changing the dimensioning defaults.

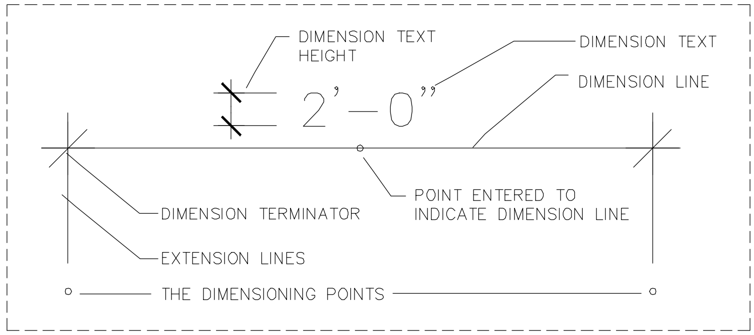

Fig. 3.12: *The components of a dimension.*

The following are the common methods for drawing dimensions:

- Drawing horizontal and vertical dimensions
- Dimensioning from a base line
- Dimensioning arcs and circles
- Drawing dimensions parallel to an object
- Dimensioning angles

Drawing Horizontal and Vertical Dimensions

You can draw horizontal or vertical dimensions between any two points. Even if the points you indicate are not aligned horizontally or vertically, the computer automatically calculates the horizontal or vertical distance between the points and draws the dimension.

Diagrams A and B (Fig. 3.13) illustrate some examples of horizontal and vertical dimensions. The dots shown in the diagram are the dimensioning points that were entered to draw the dimensions.

Most CADD programs allow you to draw a string of dimensions. When you have finished drawing the first dimension, the next dimension can be drawn just by indicating the next dimensioning point. It is always better to start dimensioning from one corner and continue towards another.

Dimensioning from a Baseline

You can draw a series of dimensions measured from a single reference point. This mode of measurement is often referred to as baseline, benchmark or datum. To draw dimensions from a baseline, you need to draw the first dimension by indicating the dimensioning points and the dimension line. The rest of the dimensions are automatically drawn measured from the baseline, that is, the first point you entered. Diagram C (Fig. 3.13) illustrates some examples of benchmark dimensions that are taken from the left corner of the diagram.

Dimensioning Arcs and Circles

To dimension arcs and circles, all you need to do is indicate the arc or circle and a dimension showing the radius or diameter value is drawn. All necessary annotations are automatically drawn. You can choose from a number of annotation styles available.

Drawing Dimensions Parallel to an Object

You can draw dimensions parallel to any two indicated points. It gives the exact distance between the indicated points. This mode of dimensioning is quite helpful for dimensioning odd-shaped diagrams. The dimension line is automatically drawn parallel to the indicated points. Diagram D (Fig. 3.13) shows two examples of drawing dimensions parallel to lines.

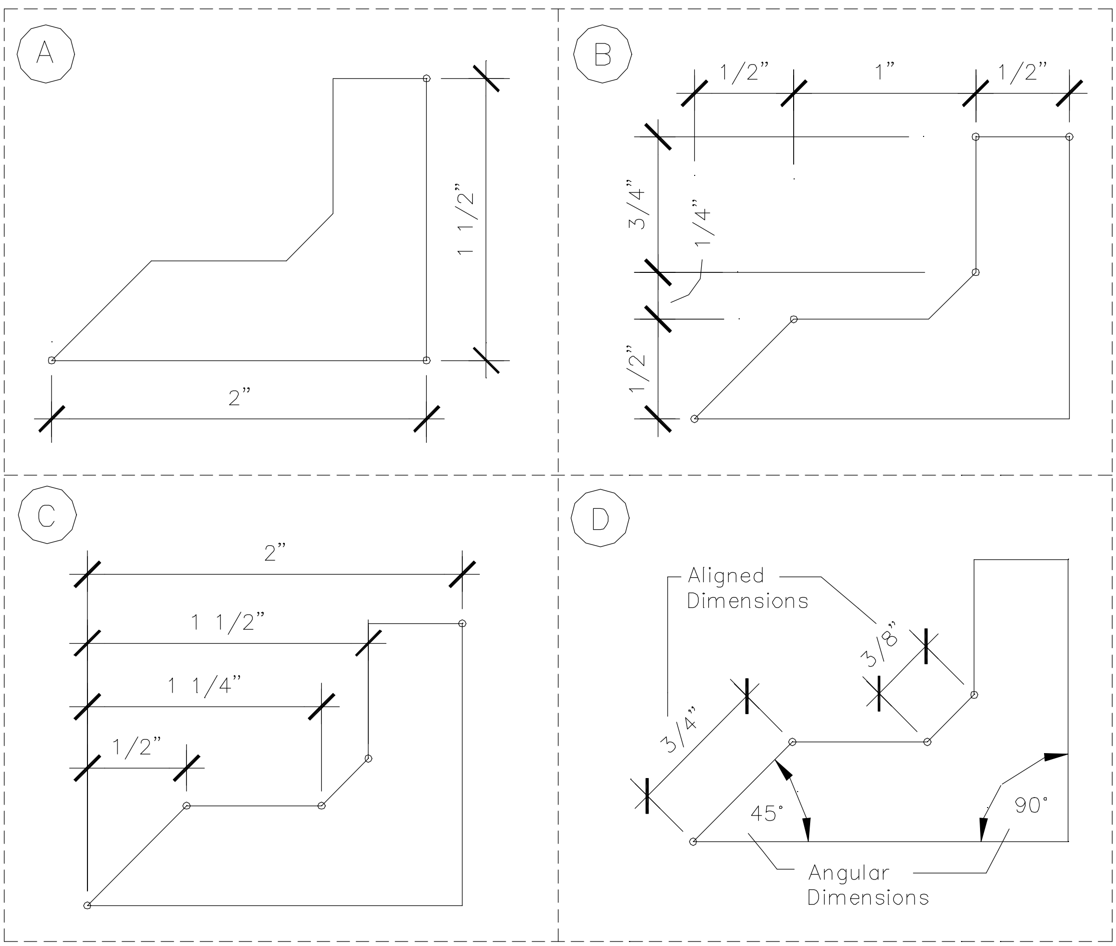

Fig. 3.13: *Drawing dimensions.*

Dimensioning Angles

CADD allows you to dimension an angle formed by two lines. To draw an angular dimension, you just need to indicate the two lines. The computer automatically calculates the angle between them and draws all the necessary annotations required for the dimension. Diagram D (Fig. 3.13) illustrates two examples of drawing angular dimensions.

Important Tip:

The key to drawing accurate dimensions is to have an accurate drawing and to indicate the dimensioning points accurately. When you first draw diagrams, they must be accurate so that when you indicate the dimensioning points, exact dimensions are drawn. If the diagram is inaccurate, it is reflected in the dimension values.

Writing Dimensions with Different Units

CADD enables you to draw dimensions with units commonly used in architecture and engineering drawings. The dimensions can be drawn in feet-inches or the metric system. Diagram A (Fig. 3.14) illustrates how the same dimension can be written with different units of measurement. The top dimension is written with architectural units using feet-inches. The following dimensions are written with decimal units, engineering units and fractional units.

When you start a drawing, you can specify the units that you will be using and the level of accuracy required. All dimensions are drawn according to that mode of measurement and accuracy. The level of accuracy that you should use for a drawing generally depends upon the scale of the drawing. When you are working on large-scale drawings, you may want to set a lesser degree of accuracy to avoid unnecessary fractions. When you are working with minute details, you may want to set a higher degree of accuracy.

Diagram B (Fig. 3.14) illustrates how the same dimension is written at varying levels of accuracy. The top dimension is written with an accuracy of 1/16" using architectural units; the following dimensions use accuracy set to 1/8", 1/4" and 1/2". As shown, the fractions are rounded off to the nearest fraction when the level of accuracy is changed.

Note:

You can change the dimensions from one mode of measurement to another or change their appearance using CADD's edit functions. For example, the dimensions created with feet-inch units can be easily converted into the metric system with a few simple steps. Similarly, you can edit the text size, font, and dimension terminators, etc.

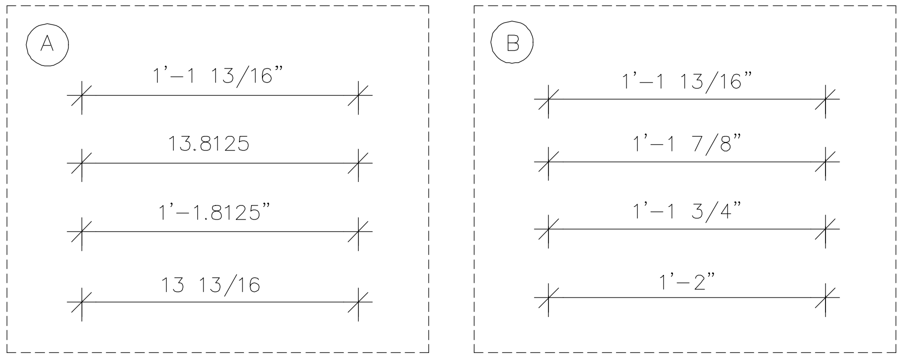

Fig. 3.14: *Drawing dimensions using different units.*

Setting Dimension Styles

CADD provides a number of options for controlling the appearance of dimensions. You can assign styles to dimension text, dimension terminators, etc. Almost all styles that are available to draw text with CADD are available to draw the dimension text as well. Diagram A (Fig 3.15) illustrates examples of dimensions written with different text styles. It is a good idea to use the same size and style for dimension text as for other descriptive text used in a drawing. This gives a uniform look to drawings.

You can choose from a number of dimension terminators. Diagram B (Fig 3.15) illustrates examples of common terminators used in CADD. You can also specify a number of other parameters, such as the size of terminators, thickness of dimension lines, length of the extension lines, distance of text from the extension line and distance of extension lines from dimension points.

CADD allows you to pre-define dimension styles that contain information about dimensions. You can specify all the dimensioning parameters, such as text height, width, ratio, font and terminators in a style. When you need to write a dimension with a specific style, simply select that dimension style from the menu.

Important Tips:

- Most CADD programs allow you to add special annotations that appear with specific dimensions. You can add prefixes or suffixes such as “(+/-)”, "field verify", "typical", "hold this dimension" and "not to scale". These annotations are commonly used in engineering drawings for specific drawing conditions.

- When CADD diagrams are reduced or enlarged using editing commands, the dimension values are updated automatically. This enhanced dimensioning feature is called “associative dimensioning”. Previously, associative dimensioning was available only in advanced CADD programs, but it is now used in almost all CADD programs.

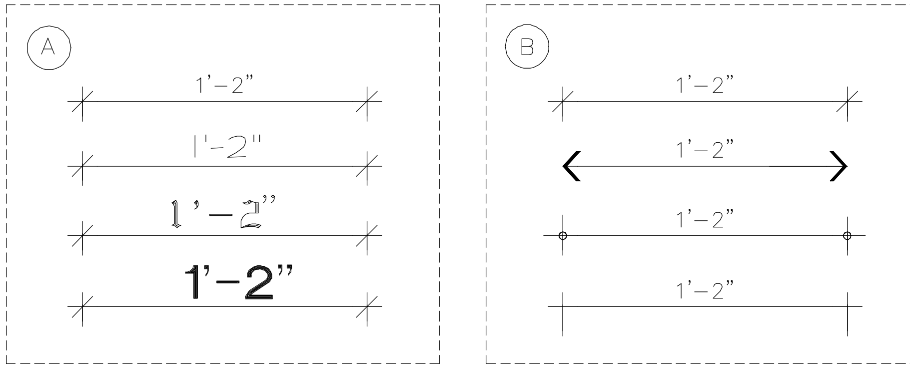

Fig. 3.15: *Drawing dimensions using styles.*

Adding Hatch Patterns to Drawings

The look of CADD drawings can be enhanced with the hatch patterns available in CADD. The patterns can be used to emphasize portions of the drawing and to represent various materials, finishes, and spaces. Several ready-made patterns are available in CADD that can be instantly added to drawings. Fig. 3.16 illustrates some common examples of hatch patterns.

Hatch patterns are quite easy to draw. You don't need to draw each element of a pattern one by one. You just need to specify an area where the pattern is to be drawn by selecting all the drawing objects that surround the area. The selected objects must enclose the area completely, like a closed polygon. When the area is enclosed, a list of available patterns is displayed. Select a pattern, and the specified area is filled.

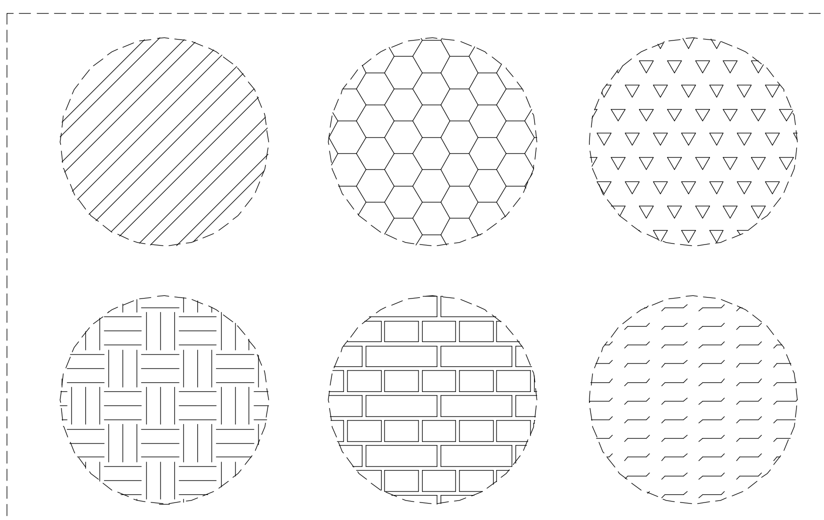

Fig. 3.16: *A number of ready-made hatching patterns are available in CADD.*

Important Tip:

A number of hatch patterns are available through independent CADD vendors. These pattern libraries are designed to help professionals in all fields. Many CADD programs allow you to create custom patterns as well. You can modify the existing patterns or define new patterns by adjusting the elements of a pattern.

Drawing Symbols

Symbols and polygons provide a convenient way to draw geometrical shapes. You may compare this function with the multi-purpose templates commonly used on a drawing board. To draw a geometrical shape, such as a pentagon or hexagon, select an appropriate symbol from the menu, specify the size of the symbol, and it is drawn at the indicated point. Some examples of symbols are shown in Diagram A (Fig.3.17).

Drawing Arrows

Arrows (or leaders) in a drawing are commonly used to indicate which note or specification relates to which portion of the drawing, or to specify a direction for any reason. There are several arrow styles available in CADD programs. Some common examples are shown in Diagram B (Fig. 3.17). You can choose from simple two-point arrows to arrows passing through a number of points, and from simple to fancy arrow styles. To draw an arrow, you need to indicate the points through which the arrow will pass.

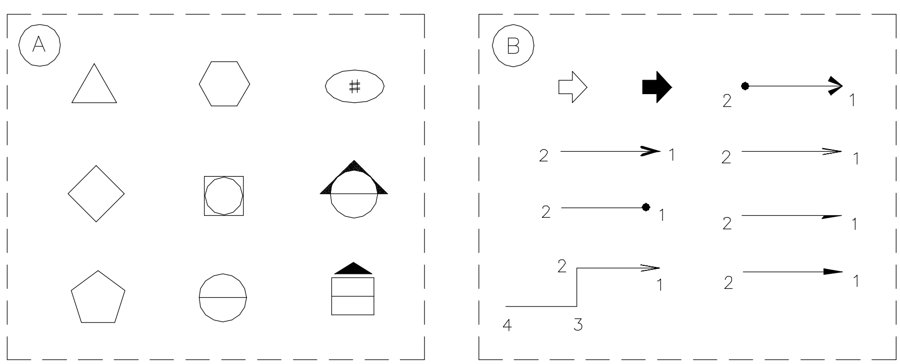

Fig. 3.17: *Drawing symbols and arrows.*